

# KUNAL M SHENDE

Embedded Systems Engineer | IoT Security Researcher | Cybersecurity Analyst

Phone: +91 8767755741 | Email: [krunalshende1436@gmail.com](mailto:krunalshende1436@gmail.com) | Location: Nagpur, Maharashtra, India

LinkedIn: [linkedin.com/in/kunal-shende04](https://linkedin.com/in/kunal-shende04) | GitHub: [github.com/KunalShende1436](https://github.com/KunalShende1436) | Portfolio: [kunalshende.me](https://kunalshende.me)

## PROFESSIONAL SUMMARY

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Computer Science undergraduate with 2+ years of hands-on experience in embedded systems, IoT hardware security, and web development. Designed ZeroTrace, an ESP32-S3-based cybersecurity audit device featuring 4 security modules and real-time RF spectrum analysis, which won 2nd Prize among 50+ teams at a national-level competition. Filed an Indian Design Patent (Application No. 499678-001). Proficient in vulnerability assessment, penetration testing, firmware development, and deploying scalable web platforms with Python Flask and cloud services.

## TECHNICAL SKILLS

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**Programming Languages:** C, Embedded C, C++, Python, TypeScript, Bash

**IoT and Embedded Hardware:** ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

**Web Development:** HTML5, CSS3, Bootstrap, Python Flask, WebSockets, Firebase

**Databases:** MySQL, Supabase, Firebase Firestore

**Cybersecurity Tools:** Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

**DevOps and Cloud:** Git, GitHub, Docker, Vercel, Google Cloud Platform, VS Code, Postman

## WORK EXPERIENCE

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### Independent Security Researcher | Freelance | January 2024 - Present

- Architected 3 secure IoT prototypes on ESP32-S3 and ATmega328P platforms, reducing firmware attack surfaces by 40% through memory-safe coding practices.
- Discovered 15+ critical vulnerabilities across 10 controlled environments by applying OWASP and NIST assessment frameworks.
- Mitigated 25% of network security anomalies through packet-level traffic analysis and proactive system hardening across 5 IoT deployments.

### Web Exploit Hunting Intern | EduSkills and AICTE | June 2026 - August 2026

- Detected 12 security flaws including SQL injection and cross-site scripting across 8 web applications during structured penetration testing exercises.
- Accelerated remediation cycles by 30% through compiling vulnerability reports with severity classification and proof-of-concept documentation.

### Cybersecurity Intern | Palo Alto Networks and EduSkills | January 2026 - March 2026

- Analyzed 500+ network packet captures and firewall rule sets, strengthening endpoint protection across 3 multi-cloud environments.
- Earned 4 industry certifications covering Security Operations, Network Security, Cybersecurity Fundamentals, and Cloud Security from Palo Alto Networks.

### Java Full Stack Developer Virtual Intern | EduSkills and AICTE | October 2025 - December 2025

- Completed 12-week structured coursework covering full-stack application development, relational database integration, and backend service design.

### Cloud Virtual Intern | AWS Academy | April 2025 - June 2025

- Studied 10+ core AWS infrastructure services including EC2, S3, VPC, and IAM through hands-on lab exercises over 8 weeks.

### Android Developer Virtual Intern | Google for Developers | January 2025 - March 2025

- Explored 8 Android development modules covering Material Design principles, activity lifecycle management, and local SQLite data persistence.

## PROJECTS

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**ZeroTrace - Cybersecurity Audit Device** | ESP32-S3, FreeRTOS, C++, USB-HID | [\[GitHub Repo\]](#) | 2026

- Engineered an all-in-one embedded audit companion integrating 4 security modules: RF scanning, Wi-Fi auditing, RSSI foxhunting, and USB-HID injection.
- Optimized display rendering to 60 FPS by pinning FreeRTOS tasks across dual CPU cores, cutting frame latency by 35%.
- Secured 2nd Prize at TECHTANTRA 2K26 national competition among 50+ teams and filed Design Patent (Application No. 499678-001).

#### **Sentinel - HID Attack Detection Framework** | *Python, Windows API, WMI* | 2026

- Developed a real-time USB threat detection system that neutralizes rogue HID devices within a sub-10-millisecond response window.
- Intercepted 100% of simulated BadUSB attack vectors through automated forensic snapshotting in controlled test scenarios.

#### **MicroFlasher - Web Serial Flashing Platform** | *TypeScript, Web Serial API* | 2026

- Deployed a zero-install browser-based firmware flasher, eliminating native toolchain setup and reducing developer onboarding time by 70%.

#### **Autonomous Line-Follower Robot** | *ATmega328P, L298N, IR Sensors* | [\[GitHub Repo\]](#) | 2025

- Constructed a differential-drive robot with hardware PWM calibration at 7.8 kHz, eliminating motor stalling and reducing audible noise by 90%.
- Calibrated a proportional control algorithm achieving 95% path-tracking accuracy on sharp 90-degree corners at competition speed.

### **ACHIEVEMENTS**

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- Secured 2nd Prize at TECHTANTRA 2K26 National Level Technical Competition among 50+ participating teams for the ZeroTrace project.
- Filed Indian Design Patent (Application No. 499678-001, CBR No. 209757) in April 2026 for ZeroTrace hardware design.

### **EDUCATION**

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**Bachelor of Technology in Computer Science and Engineering** | *S.B. Jain Institute of Technology, Nagpur* | August 2023 - June 2027  
*Specialization: IoT Architecture, Cybersecurity, Embedded Systems*

**Higher Secondary Certificate (HSC), Science** | *Municipal Junior College, Kalmeshwar* | June 2021 - March 2023

**Secondary School Certificate (SSC), Distinction** | *Municipal Junior College, Kalmeshwar* | June 2019 - March 2021

### **CERTIFICATIONS**

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**Cyber Security Workshop** - *Crow Security and IIT Bombay*, 2025-2026

**Ethical Hacking and Penetration Testing** - *NASSCOM FutureSkills Prime*, 2025-2026

**Security Operations Fundamentals** - *Palo Alto Networks*, March 2026

**Network Security Fundamentals** - *Palo Alto Networks*, March 2026

**Cybersecurity Fundamentals** - *Palo Alto Networks*, March 2026

**Cloud Security Fundamentals** - *Palo Alto Networks*, March 2026

**Python Basic and Problem Solving** - *HackerRank*, 2025

**AWS Cloud Computing Fundamentals** - *AWS Academy*, 2025

**German Language Proficiency Level 1** - *FLOA*, December 2025

### **LANGUAGES**

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English (Professional Proficiency) | Hindi (Native) | Marathi (Native) | German (Beginner - Level 1)